

Oncology: Kidney Disease in Cancer Patients

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CANCER AND KIDNEY DISEASE

Malignancy is commonly associated with acute and chronic kidney disease (Box 67.1), which in turn increases morbidity and mortality in cancer patients.¹⁻³ Rapid advancements in oncology over the past 2 decades have turned many previously fatal cancers into chronic diseases. This relates mainly to recognition of gene mutations in various malignancies that are prime targets of anticancer therapies and using drugs to leverage the immune system for killing tumor cells. Cancer therapy with conventional chemotherapeutic agents, as well as targeted agents, immunotherapies, cytotoxic T-cell therapies, and stem cell transplantation, underpins this transition. However, these positives come with a price, including acute and chronic kidney disease. Nephrologists have been inundated with information on the nephrotoxicity of new drugs and appropriate dosing of anticancer drugs in chronic kidney disease (CKD) and end-stage kidney disease (ESKD) patients, as well as the need to manage CKD in cancer survivors. An increased knowledge gap for many clinicians providing kidney care for these patients and the need for targeted education in this area has fueled the birth of onconeurology, an area of subspecialization within nephrology.

ACUTE KIDNEY INJURY

Acute kidney injury (AKI) is common in cancer patients, particularly in those who are hospitalized or admitted to the intensive care unit (ICU). In these settings, AKI is associated with increased lengths of stay and higher costs. The 1-year risk for AKI as defined by the *risk, injury, failure, loss of kidney function, and end-stage kidney disease (RIFLE)* in a large Danish study of incident cancer patients was 17.5%.¹ The 1-year risk for more severe AKI, defined as the *injury and failure* categories, was 8.8% and 4.5%, respectively. Moreover, the 5-year risks for the *risk, injury, and failure* AKI categories were even higher, at 27%, 14.6%, and 7.6%, respectively. The incidence of AKI was highest in patients with renal cell cancer, liver cancer, multiple myeloma, and leukemia. Among 9613 cancer patients with any AKI stage, 5.1% required renal replacement therapy (RRT) within 1 year of AKI onset. Unsurprisingly, older patients were most heavily represented in this analysis. In patients admitted to a comprehensive cancer center, AKI was 12%, nearly threefold higher than most noncancer settings, and was associated with worse clinical outcomes.⁴ Among those with acute kidney injury, 45% of AKI developed in the first 48 hours and the remainder developed thereafter.⁴

Critically ill cancer patients are at even higher risk of AKI than critically ill patients without cancer. Of 288 patients admitted to a cancer-focused ICU from 2006 to 2008, 54.1% developed AKI.⁵ The *risk* category occurred in 33.3% of patients, and the *injury* and *failure* categories each developed in 10.4% of patients. In addition to an increased risk for AKI, critically ill cancer patients have a higher incidence of AKI requiring RRT compared with patients without cancer. Depending on the AKI definition employed and the underlying case mix, 13% to 42% of critically ill patients with cancer may develop AKI, and 8% to 60% require RRT.⁶ Those with hematologic malignancies, myeloma, and septic shock are at highest risk to develop AKI. A study of 429 cancer patients admitted to the ICU demonstrated that AKI was an independent risk factor for both ICU and 28-day mortality. In addition, developing AKI after 24 hours in ICU was associated with higher mortality than admission AKI.⁷

Hospital-acquired AKI in cancer patients can be due to prerenal, intrarenal, and postrenal causes (see Chapter 70). Risk factors for AKI in cancer patients include typical comorbidities such as underlying diabetes mellitus and CKD, as well as exposure to radiocontrast, chemotherapy, nonsteroidal antiinflammatory drugs (NSAIDs), and antibiotics.⁴ It is therefore critical to identify at-risk cancer patients to allow prophylactic measures. This will allow clinicians to avert AKI and improve clinical outcomes for this fragile group.⁴

Prerenal Acute Kidney Injury

Nearly one-third of AKI in cancer patients is prerenal and results from volume depletion from chemotherapy-induced nausea, vomiting, and diarrhea; chemotherapy-associated hypoalbuminemia contributes to the reduction in effective plasma volume and to compromised renal perfusion. Extracellular volume depletion as a result of third spacing, as in malignant ascites or pleural effusion or insensible losses from neutropenic fever, also can lead to prerenal AKI. Abdominal compartment syndrome from tense ascites can lead to AKI from renal underperfusion.

Intrarenal Acute Kidney Injury

Intrarenal AKI is one of the most common forms of renal injury in cancer patients. The cause is often multifactorial (see Chapter 70). Several chemotherapeutic agents and other nephrotoxic medications commonly used in cancer patients may cause toxic acute tubular necrosis (ATN). Cancer patients with sepsis are also at risk to develop severe ischemic ATN.⁷ High and repetitive doses of chemotherapy,

BOX 67.1 Kidney Disease Associated With Common Cancers and Cancer Treatments

Leukemia

Common: AKI from sepsis, volume depletion, drug toxicity

Rare: infiltrative disease, GN, AKI from TLS, and leukostasis

Multiple Myeloma

Common: cast nephropathy, AKI from volume depletion and drug toxicity

Rare: amyloidosis, LCDD, Fanconi syndrome, AKI from hypercalcemia, C3 glomerulopathy, PGNMIGD

Lymphoma

Common: AKI from TLS and volume depletion

Rare: obstructive uropathy, infiltrative disease, MCD (Hodgkin lymphoma), MPGN (non-Hodgkin lymphoma)

Renal Cell Carcinoma

Common: anti-vascular endothelial growth factor toxicity, postnephrectomy CKD

Rare: obstructive uropathy, membranous GN

Lung and Head and Neck Cancer

Common: platinum toxicity

Rare: SIADH, membranous GN

Genitourinary and Gynecologic Cancers

Common: obstructive uropathy

Rare: platinum toxicity

Kidney Disease in Commonly Used Cancer Treatments

Common: AKI, tubulopathy and CKD from chemotherapeutic agents (e.g., cisplatin, ifosfamide, methotrexate), complications of conditioning regimen and HSCT, and from toxicities of targeted anticancer therapies (includes proteinuria, TMA, and hypertension), acute interstitial nephritis from immune checkpoint inhibitors

Rare: radiation nephritis, TMA, GN

Obstructive Uropathy

Common: genitourinary and gynecologic cancers

Rare: lymphoma

AKI, Acute kidney injury; CKD, chronic kidney disease; GN, glomerulonephritis; HSCT, hemopoietic stem cell transplantation; LCDD, light chain deposition disease; MCD, minimal change disease; MPGN, membranoproliferative GN; PGNMIGD, proliferative GN with monoclonal immunoglobulin deposits; SIADH, syndrome of inappropriate antidiuretic hormone; TLS, tumor lysis syndrome; TMA, thrombotic microangiopathy.

concurrent with other nephrotoxic agents, may increase AKI risk. Drug-induced acute interstitial nephritis (AIN) is a common cause of AKI in cancer patients, and exposure to the new immunotherapies and targeted therapies is increasing the prevalence of AIN.⁸

Postrenal Acute Kidney Injury

Genitourinary cancers may cause urinary tract obstruction and postrenal AKI. Cancers commonly associated with urinary tract obstruction involve the bladder, prostate, cervix, and ovary. Among these, pelvic irradiation and urogenital surgery for cancer are more likely to cause urinary obstruction. Cervical and ovarian cancers with metastatic spread involving the ureters or bladder are associated with postrenal AKI. In hematopoietic stem cell transplantation (HSCT) recipients with BK infection, ureteral obstruction is a common cause of renal

impairment, and bulky retroperitoneal adenopathy associated with lymphoma may cause obstruction.

CHRONIC KIDNEY DISEASE

Chronic kidney disease is also a complication of many cancers and their therapy, and, like AKI, it is associated with increased morbidity and mortality. Preexisting CKD from underlying comorbidities is highly prevalent in patients with various types of malignancy. Additionally, CKD may develop after multiple episodes of AKI, after nephrectomy for kidney cancer, and from nephrotoxic medications that cause glomerulosclerosis and tubulointerstitial fibrosis. The Renal Insufficiency and Anticancer Medications (IRMA) studies observed that 53% (IRMA-1) and 50% (IRMA-2) of patients with active malignancy had an estimated glomerular filtration rate (eGFR) less than 90 mL/min/1.73 m², respectively.⁹ In this group, the prevalence of stage 3 CKD was 12% in both studies, whereas stage 4 CKD was rare (0.9% and 0.7%, respectively). In an Australian cohort with various cancers ($n = 4077$), eGFR less than 60 mL/min/1.73 m² was observed in 30% of patients, and 8% had eGFR less than 45 mL/min/1.73 m².¹⁰ In a cohort of Chinese patients with cancer, eGFR 30 to 59.9 mL/min/1.73 m² was noted in 13% of 8223 patients.¹¹ The relatively common occurrence of CKD in many cancer patients has been confirmed in other studies.

The relationship between kidney disease and cancer appears to be bidirectional. Although there is an increase in CKD prevalence in cancer patients, both CKD and ESKD are risk factors for development of a number of malignancies¹ (see [Chapter 92](#)). In a retrospective cohort study of more than 1 million adults, lower eGFR was associated with an increased risk for kidney cancer after adjustment for time-updated confounders. In addition, the adjusted hazard ratio was 2.28 (95% confidence interval, 1.78–2.92) for eGFR less than 30 mL/min/1.73 m².¹² In patients with ESKD on dialysis, there is an increased risk for renal parenchymal cancer in patients with acquired renal cystic disease.

MORTALITY IN CANCER PATIENTS WITH KIDNEY DISEASE

The increased mortality associated with cancer-related AKI and CKD is related to a number of factors. The occurrence of AKI may require cessation of effective chemotherapeutic regimens, allowing unhindered tumor growth. Preexisting CKD may limit use of chemotherapeutic agents or promote underdosing of curative anticancer regimens. CKD and AKI may alter the bioavailability and/or safety profile of some anticancer drugs leading to use of suboptimal therapies. In addition, CKD and ESKD may impair immune surveillance, allowing cancers to grow and metastasize, especially when combined with insufficient drug dosing. It is also possible that anticancer agent nephrotoxicity is associated with worsening of preexisting CKD, which leads to increased all-cause mortality unrelated to cancer. A recent retrospective study of 128,489 patients from Ontario, Canada, with newly diagnosed solid cancer concluded that survival among patients with advanced kidney disease (stages 3a–5), ESKD, and transplants was inversely correlated with kidney function. In addition, receipt of all treatment modalities (systemic therapy, radiation, and palliative care) was significantly reduced in this group of patients.¹³

IMPORTANT MALIGNANCIES ASSOCIATED WITH KIDNEY DISEASE

Multiple Myeloma and Amyloidosis

Kidney involvement occurs in about 50% of patients with myeloma but varies depending on the myeloma type¹⁴ (see [Chapters 28 and 68](#)

). Monoclonal light chain (LC) overproduction is commonly associated with kidney injury, with as many as 50% of patients requiring dialysis.¹⁵ The LC type is also important; cast nephropathy and renal amyloidosis are common with λ LCs, whereas severe proximal tubular injury causing Fanconi syndrome and LC deposition disease are common with κ LCs. Cast nephropathy has been diagnosed in 41% of patients with myeloma and kidney disease, and the use of therapeutic plasma exchange or high cut-off hemodialysis has not been demonstrated to improve kidney or overall survival outcomes.¹⁶ Prior to the use of proteasome inhibitors, kidney failure in patients with myeloma was associated with reduced overall survival. Myeloma and amyloid patients' outcomes have further improved with autologous stem cell transplantation (ASCT), which is now standard care in patients who have adequate organ function. Multiple studies have demonstrated improved kidney function following ASCT.¹⁷ Importantly, myeloma-related kidney failure did not affect progression-free survival or overall survival.¹⁸ Primary systemic amyloidosis in patients with LC disease is another infrequent manifestation of myeloma (see Chapter 28). When amyloidosis complicates myeloma, it carries a grave prognosis. Kidney enlargement, gross proteinuria, and amyloid deposition are hallmarks of renal amyloidosis, which occurs in about 5% to 10% of patients with myeloma based on autopsy studies.¹⁹

Leukemia and Lymphoma

Various forms of kidney disease occur in patients with lymphoma and leukemia. Those admitted to an ICU had an AKI hazard ratio of 2.23 for AKI, compared with otherwise comparable patients without ICU admission.²⁰ In 200 patients with newly diagnosed high-grade lymphoma and leukemia, the incidence of RIFLE AKI was 68.5%, with 50% requiring RRT.²¹ Possible causes of AKI in patients with leukemia and lymphoma include prerenal processes,²² intrarenal causes such as ATN (especially from sepsis and nephrotoxic medications),²³ or post-renal causes such as mechanical obstruction from lymphadenopathy or retroperitoneal fibrosis.²⁴ Patients with leukemia or lymphoma commonly develop infiltrative disease of the kidney.²⁵ Most often, renal infiltration is asymptomatic or subclinical, as noted in autopsy studies in which 34% of patients were found to have lymphomatous infiltration of the kidneys.²⁵ Leukemic or lymphoma cell infiltration is suggested by large kidneys on imaging (Fig. 67.1A), and urinalysis shows bland urine sediment and tubular proteinuria.²⁶ When AKI does occur, it is thought to be due to tubular compression and increased interstitial pressure from the massive malignant cell infiltration (see Fig. 67.1B).²⁷ Overproduction of lysozyme, which is a freely filtered cationic protein reabsorbed in the proximal tubule, may cause ATN and/or proximal tubular injury with full-blown Fanconi syndrome.²⁸ Intrarenal leukostasis may cause AKI in patients with acute myeloid leukemia (>100,000 white blood cells). Renal leukostasis is associated with reduced kidney perfusion, blast aggregation within the renal microvasculature, and endothelial damage triggered by soluble cytokines. Emergent treatment with cytoreductive chemotherapy and leukapheresis is generally required.²⁹

ANTICANCER DRUGS AND KIDNEY DISEASE

Chemotherapeutic Agents

Oncologists generally assess kidney function before initiation of chemotherapeutic agents. Nevertheless, chemotherapy-related nephrotoxicity has become an important area for the nephrologist. Understanding renal metabolism and excretion of these drugs is critical in ensuring efficacy while also avoiding toxicity. Nephrotoxicity associated with the more common chemotherapeutic drugs includes AKI primarily from ATN and AIN, as well as other metabolic

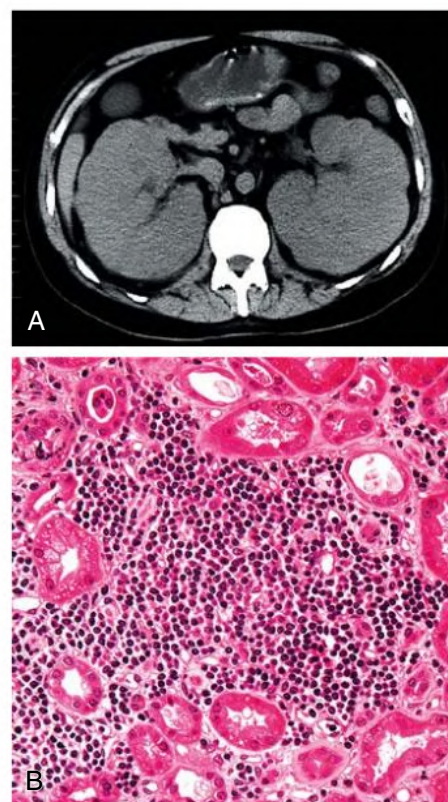


Fig. 67.1 (A) Computed tomographic scan demonstrating large kidneys from lymphomatous infiltration of the kidneys. (B) Kidney biopsy demonstrating low-grade B-cell lymphoma cells within the renal parenchyma (H&E stain; $\times 400$).

perturbations (Box 67.2). However, emergence of immunotherapies and targeted therapies has been associated with an increased incidence of various nephrotoxicities (Table 67.1).³⁰

Immunotherapies

Immune Checkpoint Inhibitors

Leveraging the immune system to destroy cancer cells has become an attractive strategy for several different malignancies. Immune checkpoint inhibitors (ICIs) impair the checkpoints that function to suppress adaptive immune response and prevent autoimmunity. T cells possess surface receptors, such as programmed death-1 protein (PD-1) and cytotoxic T-lymphocyte–associated antigen-4 (CTLA-4), which bind to their ligands on antigen-presenting cells, leading to T-cell exhaustion. This normally prevents the development of autoimmunity. Tumor survival may be enhanced in malignancies that overexpress ligands that bind these inhibitory T-cell receptors, thereby decreasing activated T-cell infiltration into the tumor microenvironment and inhibiting antitumor T-cell responses. To combat this, monoclonal antibody drugs that block ligand binding to PD-1 and CTLA-4 receptors and antibodies directed at PD-L1 were designed to facilitate T-cell rescue and restore antitumor immunity (Fig. 67.2). Ipilimumab, a fully human immunoglobulin G1 (IgG1) monoclonal antibody blocking CTLA-4; nivolumab, a fully human IgG4 antibody blocking the PD-1 receptor; and pembrolizumab, a humanized monoclonal IgG4- κ isotype antibody against PD-1, are the more common ICIs available in clinical practice. Many more have entered the clinical arena, including PD-L1 antibodies (atezolizumab, avelumab, and durvalumab). However, blocking immune checkpoints risks development of pathologic autoimmunity and end organ injury, such as AIN. Several single and multicenter studies evaluating AKI after ICIs note the incidence to

BOX 67.2 Commonly Used Cancer Therapies Associated With Kidney Toxicity**Tubular Injury****Acute Tubular Injury**

- Platinum-based agents, zoledronate, ifosfamide, mithramycin, pentostatin, imatinib, diaziquone, pemetrexed, zoledronate

Tubular Syndromes**Renal Tubular Acidosis**

- Ifosfamide, amphotericin, calcineurin inhibitors

Fanconi Syndrome

- Cisplatin, ifosfamide, azacitidine, diaziquone, imatinib, pemetrexed

Salt Wasting

- Cisplatin, azacitidine

Magnesium Wasting

- Cetuximab, cisplatin, panitumumab

Nephrogenic Diabetes Insipidus

- Cisplatin, ifosfamide, pemetrexed

Acute Interstitial Nephritis

- Sorafenib, sunitinib, checkpoint inhibitors

Crystalline Nephropathy

- Methotrexate, acyclovir, sulfa-based drugs

Syndrome of Inappropriate Antidiuretic Hormone

- Cyclophosphamide, vincristine

Glomerular Injury**Focal Segmental Glomerulosclerosis**

- IFN, pamidronate, tyrosine kinase inhibitors

Minimal Change Disease

- IFN, pamidronate

Proteinuria

- Sorafenib, sunitinib, vatalanib, axitinib

Renal Vascular Injury**Thrombotic Microangiopathy**

- Bevacizumab (anti-VEGF monoclonal antibody)

Tyrosine Kinase Inhibitors

- Sorafenib, sunitinib, imatinib

Other Agents

- Gemcitabine, mitomycin C, IFN- α , calcineurin inhibitors

IFN, Interferon; VEGF, vascular endothelial growth factor.

range from 1.4% to 4.9%.³¹⁻³³ Among a number of lesions that have been described, AIN is the most common cause of AKI.³⁴⁻³⁶ Although some patients develop rash and eosinophilia, AKI is the only consistent clinical manifestation. A small number of cases have required RRT, but most respond to drug discontinuation and steroid therapy. The mechanisms associated with ICI-induced nephrotoxicity are poorly understood, but drugs such as NSAIDs and proton pump inhibitors (PPIs) are associated with increased risk of AIN in several studies.³¹⁻³³ Immune checkpoint inhibitor (ICI)-induced loss of tolerance to these drug antigens may lead to development of AIN. Induction of autoimmune disease or reactivation of preexisting autoimmune disease in the kidney has also been demonstrated with ICI exposure and poses a further challenge in cancer therapy, as treatment of ICI-associated inflammation often requires aggressive immunosuppression, which may promote cancer progression and offset the antitumor effects.³⁷⁻³⁹

In addition, ICI therapy for cancer may increase the risk of graft loss in kidney transplant recipients, a group with high risk for certain cancers due to immunosuppression. Single center retrospective studies and multicenter cohorts note rejection rates as high as 40% with ICI therapy in organ transplant patients and more specifically in those with kidney transplants.⁴⁰ The median time from therapy to rejection ranged from 14 days to 24 days. There are no accepted guidelines for immunosuppression management during ICI use in kidney transplants. However, prioritizing cancer treatment over salvaging a kidney transplant has been the general approach.

Chimeric Antigen Receptor T Cells

Infusion of autologous ex vivo modified T lymphocytes that express chimeric antigen receptors (CARs) targeting specific tumor antigens is a more focused approach to immune destruction of tumor cells (Fig. 67.3).⁴¹ A more expansive approach is the use of CAR-expressing natural killer (NK) cells or macrophages. These therapies are effective in treating patients with refractory/relapsing leukemias and lymphomas

with potential future use for multiple myeloma and solid tumors. CAR T-cell therapy is often associated with cytokine release syndrome (CRS), that can be severe and lethal. In addition to capillary leak syndrome and other end-organ injury such as AKI, patients may progress to hemophagocytic lymphohistiocytosis, characterized by high ferritin, low fibrinogen, and cytopenias. Prerenal physiology and ATN are toxicities associated with CRS in clinical trials. A recent single-center study of 78 patients receiving CAR T-cell therapy noted CRS in 85% of the patients, of which 19% developed AKI, split between prerenal and ATN.⁴² In addition to AKI and a high mortality, electrolyte abnormalities were also reported, including hypophosphatemia (75%), hypokalemia (56%), and hyponatremia (51%). Because CRS is primarily driven by interleukin 6 (IL-6), the anti-IL-6 antibody tocilizumab has proven effective in reversing this life-threatening complication.

Interferon Therapy

Treatment with interferon (IFN) is associated with proteinuria secondary to glomerular injury that is primarily a podocytopathy.⁴³ Minimal change disease (MCD) was first described, but more recent reports describe collapsing and noncollapsing focal segmental glomerulosclerosis (FSGS). Nephrotic-range proteinuria and AKI are observed within weeks of commencing IFN therapy. Although proteinuria declines with IFN discontinuation, complete resolution is more common with MCD than with FSGS. Corticosteroids are associated with complete remission in fewer than one-third of patients with IFN-associated FSGS; however, an attempt at corticosteroid therapy is still warranted with rapid taper if no response.

Targeted Therapies**Antiangiogenesis Therapy**

The discovery that vascular endothelial growth factor (VEGF) is an important mediator of tumor growth and angiogenesis led to the development of drugs targeting this pathway for cancer therapy.

TABLE 67.1 Kidney Toxicities Associated With Targeted Therapies

Drug Class	Renal Excretion	Most Frequent Adverse Kidney Events	EVIDENCE FOR DOSE REDUCTION in CKD		
			Mild/Moderate ^a	Severe ^b	Dialysis
VEGF/VEGFR-Targeting Agents					
Bevacizumab	No	Hypertension, proteinuria	No	No	No
Aflibercept	No	Hypertension, proteinuria	No	No	No
Sunitinib	16%	Hypertension, proteinuria	No	No	No
Pazopanib	<4%	Hypertension, proteinuria	No	No	No
Axitinib	23%	Hypertension, proteinuria	No	No	No
Other Multikinase Inhibitors					
Sorafenib	19%	Hypertension, proteinuria, hypophosphatemia	No	No	No
Regorafenib	19%	Hypertension, proteinuria, electrolyte disorders	No	No	No
Vandetanib	25%	Hypertension, proteinuria, AKI	No	Yes	No
Imatinib	13%	More renoprotective effects, AKI described	No	No	No
mTOR Inhibitors					
Everolimus	2%	Proteinuria, AKI, electrolyte disorders	No	No; suspend if AKI	No
Temsirolimus	4.6%	Proteinuria, AKI, electrolyte disorders	No	No; suspend if AKI	No
EGFR Inhibitors					
Gefitinib	<4%	Electrolyte disorders	No	No	No
Erlotinib	<9%	Electrolyte disorders	No	No	No
Afatinib	<5%	Electrolyte disorders	No	No	No
Cetuximab	No	Hypomagnesemia, other electrolyte disorders	No	No	No
Panitumumab	No	Hypomagnesemia, other electrolyte disorders	No	No	No
B-RAF Inhibitors ± MEK Inhibitors					
Vemurafenib	1%	AKI (tubular necrosis?)	No	No	Possible (risk for arrhythmia)
Dabrafenib	23%	Hypophosphatemia (granulomatous nephritis?)	No	No	No
Trametinib	<20%	Hypertension, hyponatremia (with dabrafenib)	No	No	No
ERBB2-Targeting Agents					
Trastuzumab	No	Hypertension, AKI (with cisplatin)	No	No	No
Pertuzumab	No	No issues	No	No	No
Lapatinib	2%	No issues	No	No	No
Trastuzumab emtansine	<5%	Hypokalemia	No	No	No
Antibodies Against CTLA-4					
Ipilimumab	No	Autoimmune nephritis, (potential drug reaction with eosinophilia and systemic symptom syndrome)	No	No	No
Other Agents					
Crizotinib	No	Reduction of eGFR (tubular necrosis?), renal cysts	Possible, with caution	Possible, with caution	No
Catumaxomab	No	No issues	No	No	No

^a30–90 mL/min/1.73 m².^b<30 mL/min/1.73 m².

AKI, Acute kidney injury; CKD, chronic kidney disease; CTLA-4, cytotoxic T-lymphocyte-associated antigen-4; MEK, mitogen-activated protein kinase; mTOR, mammalian target of rapamycin complex; VEGF, vascular endothelial growth factor; VEGFR, vascular endothelial growth factor receptor.

Effect of Monoclonal Immune Checkpoint Inhibitors on T-Cell Function

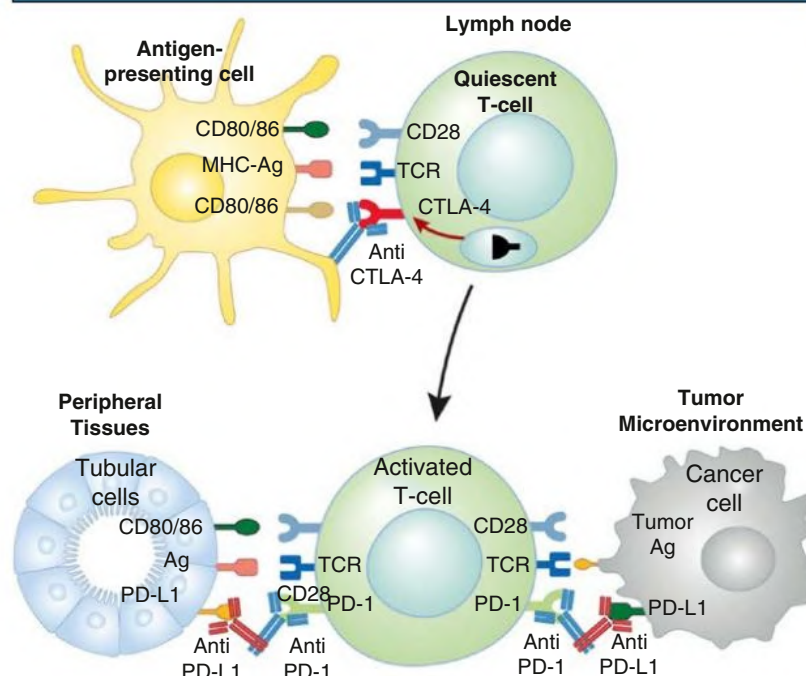


Fig. 67.2 Effect of Monoclonal Immune Checkpoint Inhibitors on T-Cell Function. In the lymph tissue, T cells are quiescent but can be activated by antigen presentation by an antigen-presenting cell. Antigen binding to the T-cell receptor (TCR) is modulated by costimulatory signals such as CD80/86 binding CD28 and coinhibitory signals such as CD80/86 binding cytotoxic T-lymphocyte-associated protein-4 (CTLA-4), which is shuttled to the cell surface upon antigen binding the TCR. Blocking the CTLA-1 receptor with monoclonal antibody allows T-cell activation, which can participate in cancer destruction. In peripheral tissues, activated T cells can be deactivated by the binding of programmed cell death ligand-1 (PD-L1; or PD-L2) to the PD-L1 receptor. Host tissues prevent autoimmune injury by deactivating T cells, whereas cancer cells prevent tumor killing through the same process. Blocking the PD-1 receptor (or blocking PD-L1) with immune checkpoint inhibitors allows T cells to remain activated and allows eradication of cancer cells. However, this can also be associated with host organ injury such as acute kidney injury. Ag, Antigen; MHC-Ag, major histocompatibility complex antigen.

However, VEGF is also an essential growth factor to maintain glomerular endothelial health. It is secreted by podocytes and traverses the basement membrane, where it binds to VEGF receptors on endothelial cells and maintains glomerular endothelial cell integrity and filtration barrier function. In addition, VEGF enhances production of local regulatory complement factors, thereby preventing pathologic endothelial injury and thrombosis.⁴⁴ Anti-VEGF drugs (e.g., the monoclonal antibody bevacizumab) and drugs that inhibit tyrosine kinase (e.g., sunitinib) both block VEGF function. This leads to glomerular endothelial cell dysfunction and filtration barrier disruption, resulting in proteinuria, thrombotic microangiopathy (TMA), hypertension, and AKI.⁴⁵ TMA may also develop from loss of VEGF-associated regulatory complement factor production. Mild and asymptomatic albuminuria is common, and heavy proteinuria occurs in less than 10% of subjects.⁴⁶ Hypertension or aggravation of preexisting hypertension is common. The reported incidence varies between 17% and 80%.⁴⁷ Development of hypertension reflects efficacy of VEGF blockade therapy, perhaps explaining why bevacizumab-induced hypertension correlates with clinical outcomes.⁴⁸ Hypertension is managed with standard antihypertensive medications, although the use of angiotensin-converting enzyme inhibitors/angiotensin receptor blockers is preferable in patients

with proteinuria. Kidney side effects that develop with anti-VEGF therapy (especially TMA) may require dose reduction or drug discontinuation.⁴⁶

B-RAF Inhibitors

Malignant melanoma frequently has a B-RAF V600 mutation that can be effectively targeted by the selective B-RAF inhibitors vemurafenib and dabrafenib. Although they are effective anticancer agents, these drugs are nephrotoxic. A GFR decline at 1 and 3 months of therapy occurred in 15 of 16 patients, which was complicated by persistent kidney injury after 8 months of follow-up. Eight patients developed AKI after vemurafenib, with ATN seen in 1 patient. AKI that was clinically suggestive of AIN was observed in 4 patients treated with vemurafenib, with 3 of 4 patients recovering kidney function after drug discontinuation, although no biopsy data were available.⁴⁹ Nearly 60% of 74 patients treated with vemurafenib developed AKI, primarily Kidney Disease: Improving Global Outcomes (KDIGO) stage 1, within 3 months of drug exposure.⁵⁰ Kidney biopsy revealed tubulointerstitial injury in 2 patients. Kidney function recovered within 3 months of B-RAF discontinuation. The combination of dabrafenib and trametinib was shown to cause AKI in 21% of 199 patients in a retrospective cohort study.⁵¹ Although the mechanism of kidney injury is unknown,

Creation of CAR T Cells

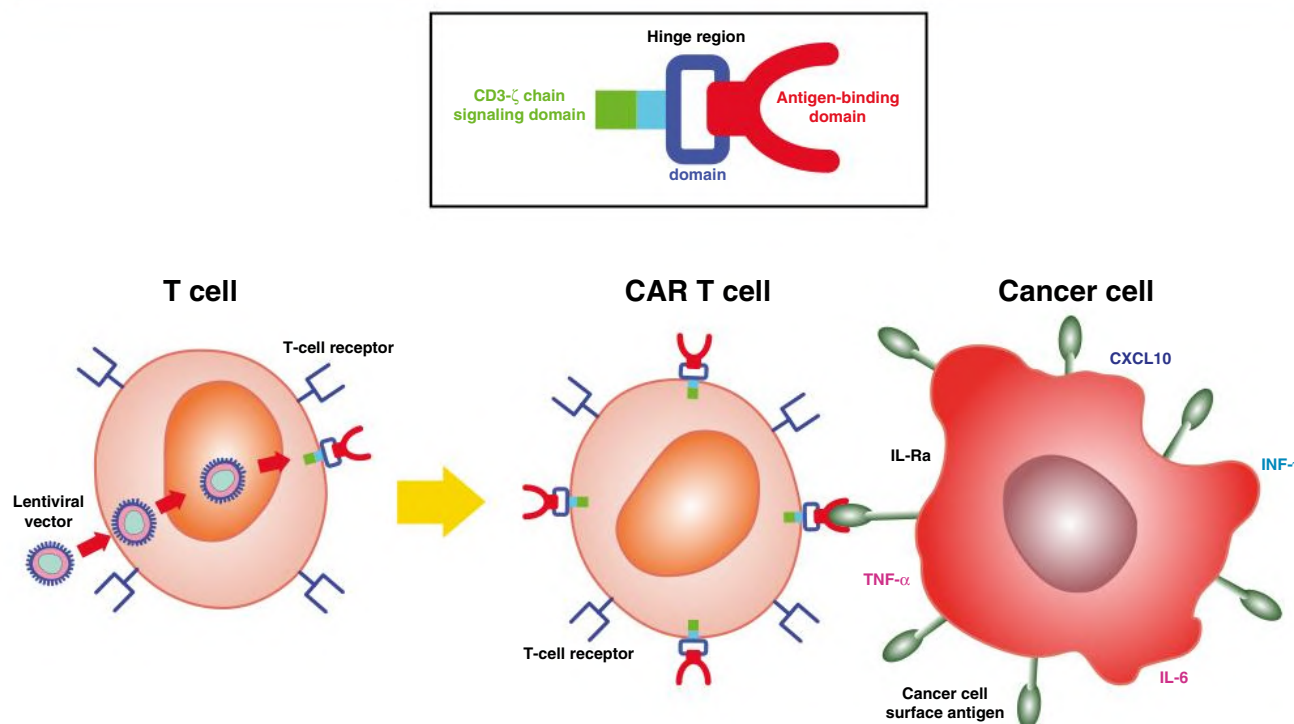


Fig. 67.3 The creation of chimeric antigen receptor (CAR) T cells involves harvesting T cells from patients and subsequently genetically modifying them using a lentiviral vector to place an antigen-binding domain, which recognizes tumor antigen. The antigen-binding receptor is linked to an intracellular costimulatory domain (CD28 or 4-1BB) and CD3- ζ signaling domain to amplify the immune response against tumor cells. Chimeric antigen receptor T cells engage tumor antigen by using extracellular antigen receptors, which are linked to intracellular costimulatory and signaling domains to amplify the immune response against tumor cells. Proinflammatory cytokines and chemokines are produced, which participate in eradication of cancer cells. *CXCL10*, C-X-C motif chemokine ligand 10; *IL-Ra*, interleukin-1 receptor antagonist; *IL-6*, interleukin-6; *INF- γ* , interferon- γ ; *TNF- α* , tumor necrosis factor- α .

these drugs may interfere with the downstream mitogen-activated protein kinase pathway, increasing susceptibility to ischemic tubular injury. Therapy is supportive along with drug discontinuation.

Anaplastic Lymphoma Kinase Inhibitors

Anaplastic lymphoma kinase 1 (ALK-1) is a member of the insulin receptor tyrosine kinase family, for which small molecule inhibitors include crizotinib and ceritinib. Crizotinib is an effective agent for treatment of advanced ALK-positive non-small cell lung cancer and appears to be associated with two major issues: AKI and an increased risk for the development and progression of renal cysts.⁵² In addition, this agent is also associated with development of peripheral edema and, rarely, electrolyte disorders. Crizotinib is associated with true and pseudo-AKI, the latter due to drug-induced reduced tubular creatinine secretion. If true AKI, temporary discontinuation of drug is recommended. Rechallenge with drug is possible if full recovery of kidney function has occurred.

METABOLIC COMPLICATIONS

Tumor Lysis Syndrome

High-grade lymphomas, acute leukemias, and other rapidly proliferating tumors are prone to tumor lysis syndrome (TLS). Among these, patients with Burkitt lymphoma are particularly at risk. Volume depletion, hypotension, large primary or metastatic tumor load, preexisting CKD, and concurrent exposure to potential nephrotoxic agents are common risk factors for TLS-induced AKI. Cell lysis leads to hyperkalemia,

hyperkalemia, hyperphosphatemia, and acidosis causing AKI, primarily from urate crystal deposition in the tubules. Hyperuricemia may also play a role in TLS-associated AKI independent of crystals, possibly as a result of inflammation and endothelial dysfunction.⁵³ Myocardial deposition of calcium-phosphate crystals may contribute to the development of life-threatening arrhythmias. TLS prevention in high-risk patients involves hydration to maintain a high urine output (>100 mL/h) and allopurinol or rasburicase to reduce hyperuricemia. Rasburicase, which degrades uric acid to water-soluble allantoin, effectively lowers uric acid levels but has not been shown to be superior to allopurinol for prevention of TLS-associated AKI.⁵⁴ Early dialysis should be considered when progressive AKI develops in patients with TLS who will be unable to handle the excessive cellular contents (e.g., potassium, phosphate) to prevent life-threatening electrolyte abnormalities. Furthermore, accumulation of tumor lysis products can aggravate renal injury. In patients with severe TLS and oliguric AKI, continuous RRT may be required to control hyperkalemia and persistent hyperphosphatemia.

Common Electrolyte Disorders

Although electrolyte abnormalities are common in cancer patients, factitious laboratory findings may occur. Paraproteins, especially IgM, may cause assay interferences with several tests that are mostly due to their physical interferences with reagents, reactions, or both.⁵⁵ Pseudohyperkalemia secondary to in vitro release of potassium from cells may occur in patients with acute leukemia, chronic lymphocytic leukemia, and chronic myeloproliferative disorders. When factitious

values are suspected, careful collection and prompt analysis of repeat samples are required.

Hyponatremia is the most common electrolyte abnormality and affects nearly 50% of hospitalized cancer patients.^{56,57} Hyponatremia is frequently due to the syndrome of inappropriate antidiuretic hormone (SIADH), which occurs with many malignancies, including lung cancers, head and neck cancers, and some hematologic cancers.⁵⁸ Nausea, pain, and analgesics such as morphine and its derivatives, antidepressants, and several chemotherapeutic agents may cause hyponatremia either by increasing arginine vasopressin secretion or enhancing its effect. Although SIADH is common, one-third of cancer cases develop hyponatremia from volume depletion, which is responsive to normal saline repletion. Treatment with fluid restriction is difficult in cancer patients with SIADH, especially in subjects receiving chemotherapy or HSCT, because these procedures require high intravenous fluid intake. Oral sodium chloride (titrated to target sodium correction) and loop diuretics are sometimes effective in those not receiving large intravenous fluid volumes, whereas vasopressin V2 receptor antagonists may be an option to correct hyponatremia in such patients.⁵⁸ Hypernatremia is much less frequent compared with hyponatremia (3% vs. 47%) but is associated with higher mortality and longer hospital stays.⁵⁹ Most hypernatremia occurs in the hospital and is observed primarily in critically ill leukemia and HSCT patients who are receiving loop diuretics.

Other common electrolyte abnormalities in cancer patients (especially those who are receiving chemotherapy) include hypokalemia, hypomagnesemia, and hypophosphatemia. Nausea/vomiting, diarrhea, and poor nutrition from chemotherapy account for many of these abnormalities. However, severe abnormalities are usually due to drug-induced Fanconi syndrome. Ifosfamide and cisplatin are the most common agents, but κ LC myeloma also can cause this syndrome. Cetuximab causes hypomagnesemia, which occasionally can be severe. In health, epidermal growth factor (EGF) binds its receptor (EGFR) and stimulates distal tubular magnesium reabsorption. Cetuximab (EGFR antibody) competes with EGF for its receptor, thereby inhibiting normal reabsorption of luminal magnesium and causing hypermagnesuria.

Hypercalcemia is common in patients with advanced cancer with bone metastasis and heralds a poor prognosis (see [Chapter 11](#)). Hypercalcemia is frequently associated with multiple myeloma and can be the presenting feature. Several mechanisms may underlie tumoral hypercalcemia such as elevated parathyroid hormone–related peptide levels associated with lung cancer, ectopic vitamin D in certain lymphomas, or tumor resorption of bones as in advanced cancer (see [Chapter 11](#)). Treatment of hypercalcemia mandates vigorous intravenous fluid administration in those who are volume depleted ([Box 67.3](#)). Although corticosteroids (for myeloma and vitamin D–related hypercalcemia) and calcitonin are useful, persistent hypercalcemia often requires bisphosphonate therapy for long-term control. Rarely, hemodialysis with low calcium dialysate alone is sufficient to control severe hypercalcemia in symptomatic patients. Use of bisphosphonates, especially zoledronate and pamidronate, can be complicated by AKI and collapsing FSGS, respectively. Ibandronate and denosumab are less nephrotoxic options.⁶⁰

CANCER-RELATED GLOMERULONEPHRITIS

Cancer-related glomerulonephritis is a relatively rare but important entity in patients with certain malignancies. Proteinuria or (rarely) nephrotic syndrome can be the manifesting feature of cancer. Membranous nephropathy (MN) (see [Chapter 21](#)) is the most common glomerular pathology associated with solid tumors. In

BOX 67.3 Management of Cancer-Related Hypercalcemia

1. General measures: IV fluids followed by loop diuretics if hypervolemia.
2. Administer calcitonin (4 U/kg SQ every 6–12 h based on calcium response), corticosteroids (prednisone 20–40 mg/d PO depending on cause), or both for acute and short-term calcium control.
3. If patient severely symptomatic and with reduced urine output and impaired kidney function, consider dialysis with low bath calcium.
4. Start bisphosphonates (pamidronate 30–90 mg IV over 2–4 h depending on GFR; zoledronate 4 mg IV over 15 min, dose adjusted for GFR, contraindicated for GFR <30 mL/min) for longer-term hypercalcemia control (adjust for kidney dysfunction or switch to denosumab; see step 6).
5. Treat the underlying malignancy.
6. For bone resorption–associated hypercalcemia: humanized anti-RANKL antibody (denosumab 60–120 mg SQ) appears to be safer and more effective than bisphosphonates (particularly in patients with AKI or CKD).
7. For PTHrP-related hypercalcemia: humanized monoclonal antibody against human PTHrP may be more effective than bisphosphonates (under evaluation).

GFR, Glomerular filtration rate; IV, intravenous; PO, orally; PTHrP, parathyroid hormone–related peptide; RANKL, receptor activator of nuclear factor κ -B ligand; SQ, subcutaneously.

the largest series of 240 patients with biopsy-proven MN, a cancer diagnosis was noted in 10%.⁶¹ Common cancers associated with MN are lung, gastric, colon, and renal carcinomas, but several other solid tumors, such as prostate and breast cancer, are also linked ([Table 67.2](#)). In Hodgkin disease, the most common lesion is MCD (see [Chapter 18](#)), possibly related to tumor-related T-cell dysfunction. Chronic lymphocytic leukemia is commonly associated with a histologic pattern of membranoproliferative glomerulonephritis (see [Chapter 22](#)), especially in the presence of cryoglobulinemia. HSCT is associated with membranous nephropathy, and, rarely, MCD can be a manifestation of chronic graft-versus-host disease.⁶² The use of IFN- α , anti-VEGF therapies, and bisphosphonates in cancer therapy is associated with various glomerular diseases and thrombotic microangiopathy.⁶³

HEMATOPOIETIC STEM CELL TRANSPLANTATION

Among the types of HSCT, the myeloablative allogeneic form requires higher doses of irradiation, chemotherapy, or a combination and carries the highest risk for developing AKI (up to 70%). Acute kidney injury can have multiple causes in HSCT recipients and is associated with poor clinical outcomes in patients receiving any form of HSCT.⁶⁴ In addition to the usual causes of AKI (volume depletion, intravenous contrast, sepsis, and drug toxicity), there are several specific renal syndromes that should be suspected based on the length of time between HSCT and the development of AKI ([Table 67.3](#)).

CANCER THERAPY IN CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

Reduced GFR complicates the selection and administration of chemotherapy and may lead to unduly high doses and drug accumulation.⁶⁵ Careful dosing and dialysis timing are critical because overdosing can aggravate systemic toxicities, whereas underdosing can lead to ineffective cancer treatment. Common risk factors for chemotherapy-induced nephrotoxicity include volume depletion,

TABLE 67.2 Cancer-Associated Glomerular Diseases

Malignancy	Glomerular Diseases
Lung cancer	MN, ^a MCD, MPGN, IgAN, FSGS, CGN, TMA
Colon cancer	MN, ^a MCD, CGN
Stomach cancer	MN
Pancreas cancer	MN, ^a MCD, IgAN
Bladder cancer	MCD
Renal cell cancer	AAA, ^a CGN, IgAN, MCD, FSGS, MPGN
Prostate cancer	MN, ^a CGN
Breast cancer	MN, ^a FSGS, MPGN, TMA
Esophageal cancer	MPGN, ^a FSGS
Gastrointestinal stromal tumor	AAA
Gastric cancer	MPGN, ^a CGN, TMA
Spleen sarcoma	AAA
Head and neck cancer	MN, ^a IgAN
Wilms tumor	MN, ^a MPGN
Teratoma	MN
Ovarian cancer	MN, ^a MCD
Cervical cancer	MN
Endometrial cancer	MN
Tongue cancer	IgAN
Mesothelioma	MCD
Melanoma	MN, ^a MPGN
Skin cancers (basal cell, squamous cell)	MN
Pheochromocytoma	MN
Thymoma	MCD, ^a FSGS, CGN, MPGN
Hodgkin disease	MCD, ^a MN, MPGN, IgAN, FSGS, CGN, AAA, anti-GBM
Non-Hodgkin disease	MPGN, ^a MCD, MN, IgAN, FSGS
CLL	MPGN, ^a MCD, MN, FSGS, CGN
AML	MN, ^a FSGS
CML	MN, ^a MCD, MPGN
MGUS	MPGN
T-cell leukemia	FSGS

^aMost common glomerular lesion.

AAA, Secondary amyloidosis; AML, acute myelogenous leukemia; CGN, crescentic glomerulonephritis; CLL, chronic lymphocytic leukemia; CML, chronic myelogenous leukemia; FSGS, focal segmental glomerular sclerosis; GBM, glomerular basement membrane; IgAN, IgA nephropathy; MCD, minimal change disease; MGUS, monoclonal gammopathy of unclear significance; MN, membranous nephropathy; MPGN, membranoproliferative glomerulonephritis; TMA, thrombotic microangiopathy.

From Luciano RL, Brewster UC. Kidney involvement in leukemia and lymphoma. *Adv Chronic Kidney Dis.* 2014;21(1):27–35.

TABLE 67.3 Causes of Acute Kidney Injury and Other Kidney Disease in Relation to the Timing of Hemopoietic Stem Cell Transplantation

Before and hours after HSCT	Tumor lysis syndrome from conditioning regimen (rare) Systemic toxicity from conditioning regimen (e.g., volume depletion [common]) DMSO toxicity from hemolysis, acidosis, and pigment nephropathy (rare)
Days to weeks after HSCT	Prerenal AKI from volume depletion, AKI from neutropenic sepsis and drug toxicity (common) AKI related to engraftment syndrome (rare)
Weeks to months after HSCT	AKI from sepsis, volume depletion, drug and radiocontrast toxicity (common) AKI from sinusoidal obstruction syndrome of the liver, GVHD, thrombotic microangiopathy, and BK virus nephropathy (rare)
Months to years after HSCT	CKD from previous AKIs, continuous use of calcineurin inhibitors especially with GVHD, and from preexisting cancer, such as myeloma (common)

AKI, Acute kidney injury; CKD, chronic kidney disease; DMSO, dimethyl sulfoxide; GVHD, graft-versus-host disease; HSCT, hemopoietic stem cell transplantation.

hypoalbuminemia, metabolic derangements, older age, presence of diabetes and other comorbid conditions, underlying AKI or CKD, presence of sepsis, and concomitant use of other potential nephrotoxins. Therefore, optimizing kidney function in those with CKD before chemotherapy can reduce risk for AKI and chemotoxicity. Moreover, administration of platinum compounds requires dose adjustments and timing with dialysis. Other commonly used agents for cancer treatment that need dose adjustment for GFR and dialysis or that should be avoided completely are melphalan, methotrexate, pemetrexed, capecitabine, hydroxyurea, fludarabine, etoposide, irinotecan, and lenalidomide.⁶⁶ In contrast, the ICIs do not need dose adjustments for liver or kidney disease. Tumor lysis developing in CKD/ESKD patients can be particularly challenging because the reduced ability to eliminate the tumor products can be life-threatening. In this situation, institution of daily dialysis or continuous RRT may become necessary. Moreover, in CKD/ESKD patients on active cancer treatment, erythropoietin should be used judiciously because higher doses may be associated with poorer cancer outcomes.⁶⁷ A suggested approach is noted in [Fig. 67.4](#).⁶⁸

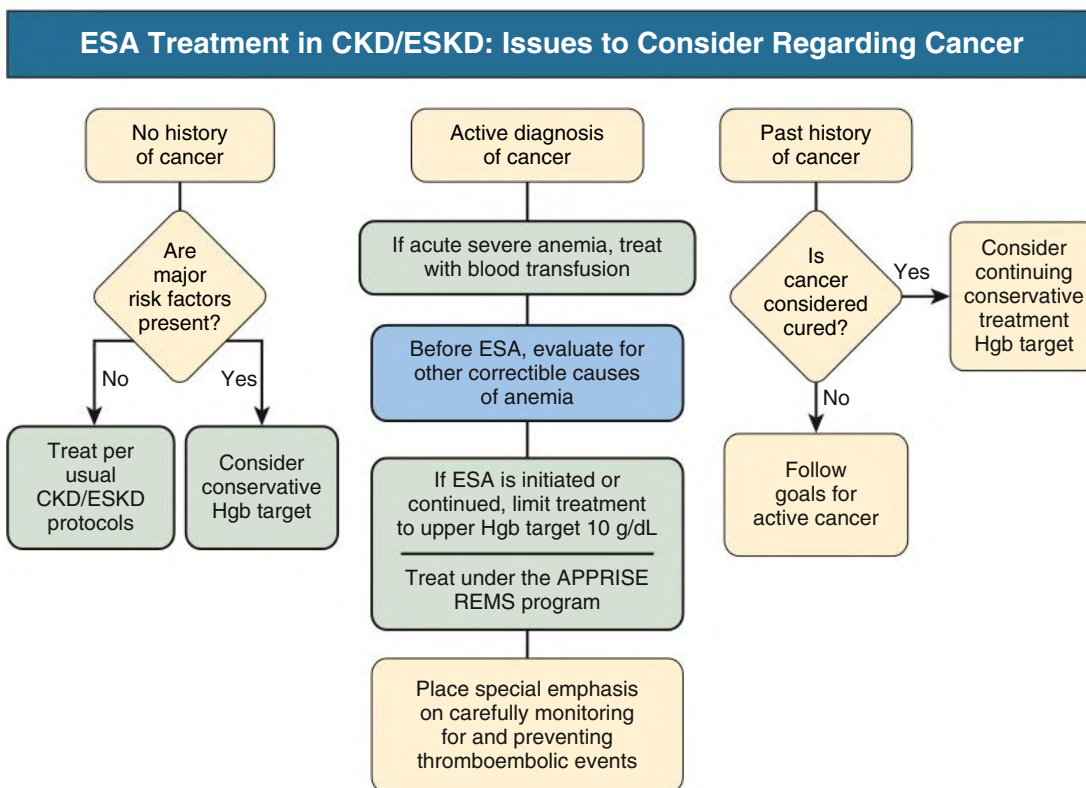


Fig. 67.4 Recommended approach to erythropoietin-stimulating agent (ESA) use in cancer patients with anemia and chronic kidney disease (CKD)/end-stage kidney disease (ESKD). *APPRISE*, Assisting Providers and Cancer Patients With Risk Information for the Safe Use of ESAs; *Hgb*, hemoglobin; *REMS*, risk evaluation and mitigation strategy.

SELF-ASSESSMENT QUESTIONS

- Which one of the following kidney lesions is most likely to be observed on kidney biopsy in a patient with multiple myeloma that develops acute kidney injury with urinalysis showing trace protein and a spot urine protein/creatinine ratio of 4.8 mg/mg?
 - Light chain–associated proximal tubulopathy
 - Light chain deposition disease
 - Primary systemic amyloidosis
 - Light chain cast nephropathy
 - Plasma cell interstitial infiltration
- A 67-year-old woman with a history of metastatic melanoma is undergoing therapy with nivolumab. After 6 weeks, the patient develops AKI (serum creatinine 3.8 mg/dL). Urinalysis reveals 1+ protein, blood, and leukocyte esterase; 3 to 5 white blood cells per high-power field (HPF), 5 to 10 red blood cells/HPF, and 3 to 5 granular casts per low-power field are noted on urine microscopy. Which one of the following lesions is *most* likely to be observed on kidney biopsy?
 - Acute tubular injury/necrosis
 - Isolated proximal tubular injury
 - Acute interstitial nephritis
 - Acute glomerulonephritis
 - Acute fibrinoid necrosis of glomerulus
- A 69-year-old woman presented with weight loss and nausea/vomiting. A computed tomography scan of the chest and abdomen demonstrated 18-cm bilateral kidneys without hydronephrosis. There was also associated retroperitoneal lymphadenopathy. Labs revealed the following: sodium 135 mEq/L; potassium 5.1 mEq/L; chloride 101 mEq/L; bicarbonate 18 mEq/L; blood urea nitrogen 59 mg/dL; creatinine 4.5 mg/dL (baseline 1.0 mg/dL); serum albumin 3.7 g/dL; calcium 11.1 mg/dL; phosphorus 7.6 mg/dL; and uric acid 11.7 mg/dL. Urinalysis showed specific gravity of 1.018, pH 5.0, and a negative dipstick examination. Urine output was 800 mL/24 h. What is the *most* likely cause of acute kidney injury in this patient?
 - Retroperitoneal lymphadenopathy with nonhydronephrotic urinary obstruction
 - Tumor lysis syndrome with acute crystalline nephropathy
 - Hypercalcemic AKI
 - Lymphomatous infiltration of the kidneys
 - Prerenal AKI from nausea and vomiting

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